

UNIT 8

ROBOTICS AND AUTOMATION



Learning Objectives

- Explain the fundamentals of robotics and automation.
- Differentiate between robots and automated systems.
- Describe the components and functions of robots.
- Understand the role of sensors and actuators in robotics.
- Explore the use of automation in various industries.
- Program simple robots using basic coding principles.
- Identify the ethical implications and future potential of robotics and automation.

Introduction to Robotics and Automation

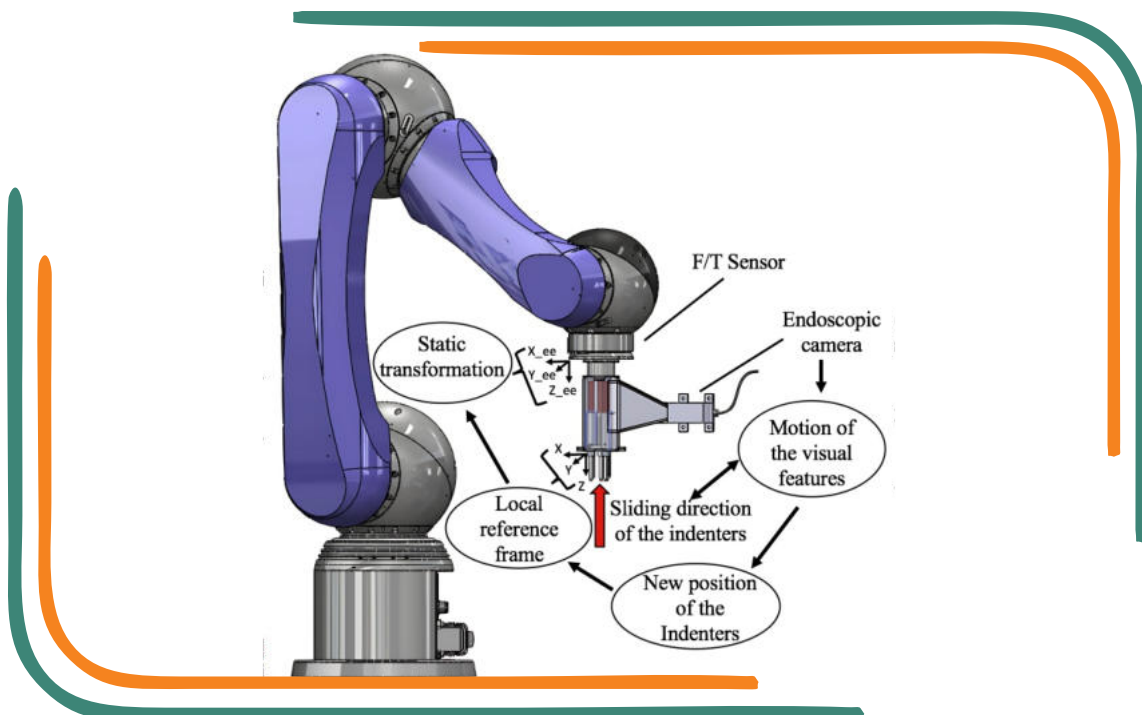
Robotics and automation are revolutionizing industries and everyday life.

While **robots** are programmable machines capable of carrying out tasks autonomously or semi-autonomously, **automation** refers to the use of technology to perform tasks with minimal human intervention.

Together, they are transforming how we manufacture goods, deliver services, and even perform household chores.

Components of a Robot

- 1. Sensors:** Detect changes in the environment (e.g., light, temperature, distance).
- 2. Actuators:** Convert electrical signals into movement (e.g., motors, servos).
- 3. Control System:** The "brain" that processes information and makes decisions.
- 4. Power Supply:** Provides energy to the robot.
- 5. End Effectors:** Tools or devices attached to the robot's arm, like grippers or welders.



Automation

- Automation is the use of machines, technology, or software to do tasks automatically without human help.
- In simple words, automation means doing work with the help of machines or computers that can follow instructions and repeat tasks without getting tired.

Examples of Automation in Daily Life:

Task	Before Automation	After Automation
Sending letters	Manually through post	Automatically through email
Cleaning floor	Done with broom manually	Automatic vacuum cleaners
Producing cars	Workers used to build every part	Robots build parts faster in factories

Where is Automation Used?

Automation is used in many areas of our daily life, homes, offices, and industries. Some common places are:

1. Home Automation

- Automatic lights and fans
- Smart TVs and voice assistants (like Alexa)
- Security systems with motion sensors

2. Office Automation

- Printers and photocopiers
- Sending emails automatically
- Scheduling online meetings

3. Industrial Automation

- Robots assembling cars in factories
- Machines packing products
- Monitoring temperature and pressure in large plants

4. Medical Automation

- Automatic machines to test blood
- Robotic surgeries
- Medicine vending machines in hospitals

5. Transport Automation

- Traffic lights working on timers
- Trains that run without drivers
- GPS navigation in vehicles

Automation in Industries

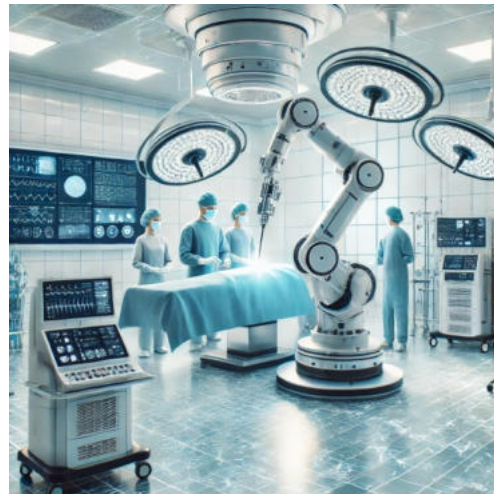
Automation increases efficiency, reduces costs, and improves product quality in various sectors.

1. Manufacturing: Automated assembly lines, robotic welding, and quality control.

2. Agriculture: Automated irrigation systems, crop monitoring drones.

3. Healthcare: Automated medication dispensers, robotic surgeries.

4. Transportation: Self-driving cars, automated public transit systems.



Advantages of Automation:

- Saves time
- Reduces human error
- Improves speed and accuracy
- Works 24/7 without breaks

Disadvantages of Automation:

- Can cause job loss for some workers
- Expensive to set up
- Needs maintenance

Programming Robots

Robots can be programmed using various languages like **Python**, **C++**, and **Blockly** (a visual, drag-and-drop coding tool suitable for beginners).

Basic Robot Programming Concepts

1. Movement Commands: Directing robots to move forward, backward, turn, etc.

2. Sensor Integration: Using sensors to avoid obstacles or follow lines.

3. Loops and Conditions: Making robots repeat tasks or make decisions based on input.

Example: Programming a robot to follow a line using infrared sensors.



Simple DIY Projects for Students

1. Line-Following Robot: Build a robot that follows a black line using light sensors.

2. Obstacle Avoidance Robot: Program a robot to detect and avoid obstacles.

3. Robotic Arm: Create a simple robotic arm to pick and place small objects.

Ethical Implications of Robotics and Automation

While robotics and automation offer numerous benefits, they also raise ethical and societal questions:

- 1. Job Displacement:** Automation can lead to unemployment in certain sectors.
- 2. Privacy Concerns:** Robots and AI may collect personal data.
- 3. Safety:** It should be ensured that robots operate safely in human environments.



EXERCISES

01: Choose the correct answer.

1. What component acts as the "brain" of a robot?

- a) Sensor
 - b) Actuator
 - c) Control System
 - d) Power Supply
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2. Which of the following is NOT an example of automation?

- a) Self-driving car
 - b) Manual painting
 - c) Robotic arm in a factory
 - d) Automated teller machine (ATM)
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3. What is the purpose of actuators in a robot?

- a) To detect environmental changes
 - b) To process data
 - c) To convert electrical signals into movement
 - d) To store energy
-

4. Which programming language is commonly used for beginner robot coding?

- a) Java
 - b) Python
 - c) Blockly
 - d) HTML
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5. Which industry uses drones for crop monitoring?

- a) Healthcare
- b) Manufacturing
- c) Agriculture
- d) Transportation

6. What ethical concern is associated with robotics?

- a) Increased creativity
- b) Job displacement
- c) Improved efficiency
- d) Enhanced safety

7. What is an example of a sensor in robotics?

- a) Motor
- b) Gripper
- c) Infrared detector
- d) Power supply

8. Which of the following is a benefit of automation?

- a) Increased human labour
 - b) Higher production costs
 - c) Consistent product quality
 - d) Slower production speed
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9. Which of the following robots would you use for precise surgical procedures?

- a) Line-following robot
- b) Robotic arm
- c) Obstacle avoidance robot
- d) Robotic vacuum cleaner

10. What is the role of end effectors in a robot?

- a) To power the robot
 - b) To store programs
 - c) To perform specific tasks like gripping
 - d) To detect obstacles
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02: Fill in the blanks.

1. _____ convert electrical signals into movement in a robot.
2. The _____ is the “brain” of a robot that processes information.
3. A robot's _____ can detect obstacles or changes in the environment.
4. _____ is the technology that performs tasks with minimal human intervention.
5. _____ are tools attached to a robot's arm for specific tasks.
6. Programming languages like _____ and C++ are used to control robots.
7. Automated _____ systems are used in agriculture for watering crops.
8. A _____ robot can detect and avoid obstacles.
9. The use of robots in healthcare includes _____ surgeries.
10. _____ concerns arise when robots collect personal data.

03: Select True or False for each statement.

	TRUE	FALSE
1. Robots always require human control to operate.	☐	☐
2. Sensors help robots interact with their environment.	☐	☐
3. Automation reduces efficiency and increases human error.	☐	☐
4. Robotics is only used in manufacturing industries.	☐	☐
5. Robotic arms can perform precise tasks like surgery.	☐	☐
6. All robots are programmed using the same language.	☐	☐
7. Automation has no impact on job markets.	☐	☐
8. Ethical concerns include privacy and safety in robotics.	☐	☐
9. Actuators are used to make decisions in a robot.	☐	☐
10. Robots can work 24/7 without needing breaks.	☐	☐

04: Answer the following questions briefly:

1. What is automation?
2. What is robotics?
3. List any two components of a robot.
4. How does automation help in factories?
5. Mention one advantage of using robots in industries.
6. Explain how robots are programmed using visual tools like Blockly.

05: Provide detailed answers of the following questions:

1. Explain what automation is and where it is used.
2. Describe the main components of a robot and explain their functions.
3. Write any four advantages and two disadvantages of automation.

Hands-On Activity:



Activity: Build a Simple Line-Following Robot

Objective: Students will assemble and program a basic robot that follows a line using light sensors.

Materials Needed: Basic robot kit (with wheels, motors, sensors), a black tape line on a white surface, and a simple programming interface (like Blockly).

Steps:

1. Assemble the robot using provided parts.
2. Connect light sensors to detect the line.
3. Write a basic program to make the robot follow the line.
4. Test and adjust the program to improve accuracy.